MAUBERT: Universal Phonetic Inductive Biases for Few-Shot Acoustic Units Discovery

MAUBERT: 用于少样本声学单元发现的通用语音归纳偏见

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Abstract

摘要

This paper introduces MAUBERT, a multilingual extension of HuBERT that leverages articulatory features for robust cross-lingual phonetic representation learning. We continue HuBERT pre-training with supervision based on a phonetic-to-articulatory feature mapping in 55 languages. Our models learn from multilingual data to predict articulatory features or phones, resulting in language-independent representations that capture multilingual phonetic properties. Through comprehensive ABX discriminability testing, we show MAUBERT models produce more context-invariant representations than state-of-the-art multilingual self-supervised learning models. Additionally, the models effectively adapt to unseen languages and casual speech with minimal self-supervised fine-tuning (10 hours of speech). This establishes an effective approach for instilling linguistic inductive biases in self-supervised speech models.

本文介绍了 MAUBERT，这是 HuBERT 的多语言扩展，利用发音特征实现鲁棒的跨语言语音表示学习。我们基于 55 种语言的语音到发音特征映射进行监督，继续 HuBERT 的预训练。我们的模型通过多语言数据学习预测发音特征或音素，从而获得与语言无关的表示，这些表示能够捕捉多语言语音特性。通过全面的 ABX 可区分性测试，我们展示了 MAUBERT 模型生成的表示比最先进的多语言自监督学习模型更具上下文不变性。此外，这些模型在极少的自监督微调（10 小时语音）下，能够有效适应未见过的语言和口语。这为在自监督语音模型中引入语言归纳偏置提供了一种有效的方法。

1 Introduction

1 引言

Is it possible to automatically discover the linguistic units of an unknown language from raw audio only? Doing so would be of great help to linguists or speech technologists working on low-resource or unwritten languages (Chen et al., 2024a; Mohamed et al., 2022; Żelasko et al., 2022; Chen et al., 2023; Zhang et al., 2021), or to cognitive modellers trying to understand how children learn their native language before learning to read and write (Kuhl, 1993; Werker et al., 2007). This question has been addressed using a variety of approaches under the Zero Resource Speech Challenge series (Versteegh et al., 2015; Dunbar et al., 2017, 2022), yielding impressive progress alongside unresolved questions.

仅凭原始音频是否可以自动发现未知语言的语言单位？这样做对于语言学家或语音技术专家在处理资源匮乏或无书写系统的语言时（Chen 等，2024a；Mohamed 等，2022；Żelasko 等，2022；Chen 等，2023；Zhang 等，2021）会非常有帮助，也可以帮助认知建模者理解儿童在学习阅读和书写之前如何学习母语（Kuhl，1993；Werker 等，2007）。这个问题已经在零资源语音挑战系列中通过多种方法进行了探讨（Versteegh 等，2015；Dunbar 等，2017, 2022），取得了令人印象深刻的进步，同时也伴随着尚未解决的问题。

Much of this progress stems from advances in self-supervised learning (SSL) techniques (Oord et al., 2019; Baevski et al., 2020; Hsu et al., 2021),

这一进展在很大程度上源于自监督学习（SSL）技术的进步（Oord 等，2019；Baevski 等，2020；Hsu 等，2021），

which have produced speech representations that capture phonetic structure better than traditional features like MFCCs or mel filterbanks. This is evidenced by improved discriminability in the learnt representation spaces: two instances of the syllable 'bit' lie closer together than one instance of 'bit' and one instance of 'bet', even across different speakers (Schatz, 2016; Schatz et al., 2013). Further evidence comes from the success of quantisation of these representations, yielding low-bitrate discrete codes suitable for training generative language models that produce novel utterances in the target language (Lakhotia et al., 2021; Borsos et al., 2023; Défossez et al., 2024; Rouard et al., 2025).

这些方法生成的语音表示能够比传统的特征（如 MFCCs 或 mel 滤波器组）更好地捕捉语音结构。这一点在学习到的表示空间中得到了体现：两个 "bit" 音节的实例比一个 "bit" 实例和一个 "bet" 实例更接近，即使在不同说话者之间也是如此（Schatz, 2016; Schatz et al., 2013）。进一步的证据来自于这些表示的量化成功，从而得到低比特率的离散码，适用于训练生成性语言模型，以生成目标语言中的新语音（Lakhotia et al., 2021; Borsos et al., 2023; Défossez et al., 2024; Rouard et al., 2025）。

However, current approaches face two limitations. First, units discovered through speech SSL do not correspond one-to-one with linguistic units like phones, syllables or words. After clustering, these units are typically shorter and more numerous than standard linguistic units: 20–40 ms long vs. 70 ms for phonemes, and N = 100–1000 vs. 30–80 for phonemes (Lavechin et al., 2025; Schatz et al., 2021). Moreover, they lack full invariance to speaker identity (de Seyssel et al., 2022; Mohamed et al., 2024) and phonetic context (Hallap et al., 2023), suggesting they capture acoustic events rather than abstract linguistic units. As a result, they produce codes with higher bitrates than phonemic transcriptions: about 100–150 bit/s versus 50–70 bit/s (Lakhotia et al., 2021; Dunbar et al., 2022). Second, current SSL algorithms require massive amounts of clean speech: Hsu et al. (2021) uses 960 hours of clean English audio, Zanon Boito et al. (2024) uses 90 k hours, and Chen et al. (2024b) uses 1 M hours. Such quantities are unavailable for low-resource languages, and notably, children acquire their language's phonetics with far less than 1000 h of much noisier input.

然而，当前的方法面临两个限制。首先，通过语音 SSL 发现的单位与语音、音节或单词等语言单位之间并非一一对应。在聚类之后，这些单位通常比标准语言单位更短且数量更多：长度为 20–40 毫秒，而音素为 70 毫秒；数量为 N = 100–1000，而音素为 30–80（Lavechin 等，2025；Schatz 等，2021）。此外，它

们缺乏对说话人身份的完全不变性（de Seyssel 等，2022；Mohamed 等，2024）以及语音环境的不变性（Hallap 等，2023），这表明它们捕捉的是声学事件，而非抽象的语言单位。因此，它们生成的代码比特率高于音素转录：大约 100–150 bit/s，而音素转录为 50–70 bit/s（Lakhotia 等，2021；Dunbar 等，2022）。其次，当前的 SSL 算法需要大量的干净语音：Hsu 等（2021）使用了 960 小时的干净英语音频，Zanon Boito 等（2024）使用了 90 千小时，Chen 等（2024b）使用了 1 百万小时。对于低资源语言而言，这样的数据量是不可用的，而且值得注意的是，儿童在远少于 1000 小时的、噪声更多的输入中就能掌握其语言的语音系统。

One avenue for improving SSL models involves pre-training universal models (Conneau et al., 2021), with recent work expanding both language

一种改进 SSL 模型的方法是预训练通用模型（Conneau 等，2021），近期的研究工作扩展了语言的覆盖面。

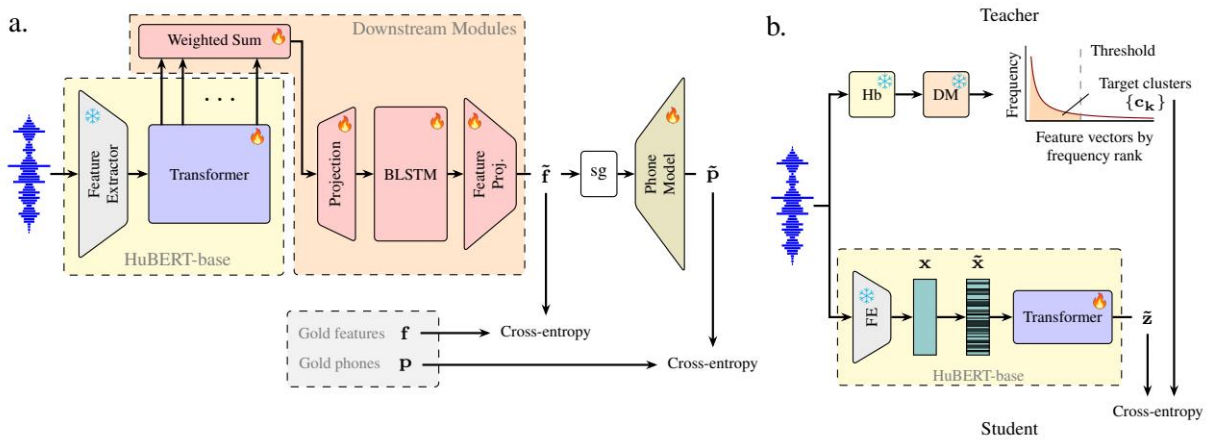

Figure 1: a. Multilingual training. MAUBERT-feat is trained to recognise ternary-valued articulatory features and phones using an encoder (HuBERT-base), downstream modules (weighted sum, up-projection, two-layer BLSTM, feature projection), and a phone model (two-layer perceptron); the feature states receive no gradients from the phone recognition loss due to the stop-gradient operator (sg). b. Self-supervised fine-tuning. Top: Offline clustering is applied to one of the layers of MAUBERT, the teacher network, on an unseen language; bottom: the MAUBERT Transformer, the student network, is then trained to predict the corresponding clusters of masked input.

图 1：a. 多语言训练。MAUBERT-feat 使用编码器（HuBERT-base）、下游模块（加权求和、上投影、两层 BLSTM、特征投影）和一个音素模型（两层感知机）来识别三值发音特征和音素；由于使用了停止梯度操作符（sg），特征状态不会从音素识别损失中接收梯度。b. 自监督微调。顶部：在未见过的语言上对 MAUBERT 的教师网络（教师网络）的一层应用离线聚类；底部：然后训练 MAUBERT Transformer（学生网络）以预测被掩码输入的相应聚类。

coverage and training data (Babu et al., 2022; Zanon Boito et al., 2024; Pratap et al., 2024; Chen et al., 2024b). Inspired by the International Phonetic Alphabet (IPA), another research direction explores how phonetically-informed target signals influence learnt representations and their cross-lingual transferability (Wang et al., 2022; Ma et al., 2023; Feng et al., 2023), suggesting that explicit phonological supervision enhances speech models' cross-lingual capabilities.

覆盖范围和训练数据（Babu 等人，2022；Zanon Boito 等人，2024；Pratap 等人，2024；Chen 等人，2024b）。受国际音标（IPA）的启发，另一个研究方向探讨了语音信息引导的目标信号如何影响学习到的表示及其跨语言迁移能力（Wang 等人，2022；Ma 等人，2023；Feng 等人，2023），表明显式的语音监督可以增强语音模型的跨语言能力。

In this paper, we explore the hypothesis that standard SSL algorithms lack strong inductive biases necessary for learning invariant speech representations from limited audio data in new languages. Following the universal pre-training and phonetically-informed research lines, we propose transforming a monolingual pre-trained SSL model (specifically, HuBERT-base trained on English) into a universal SSL model with strong inductive biases by fine-tuning it on universal IPA phonemes and features across 55 diverse languages. By directly addressing the limitations of prior work, such as the need for large datasets and the absence of explicit phonological supervision, our method provides a practical and linguistically informed solution for both speech technology and language documentation in low-resource settings. We evaluate this model, coined MAUBERT, on the ZRC2017 challenge, which presents 5 languages with less than 10h of training data (English, French, German, Mandarin, Wolof). To increase the evaluation's diversity and validity, we extend the ZRC2017 benchmark with

在本文中，我们探讨了这样一个假设：标准的 SSL 算法在从有限的音频数据中学习不变的语音表示时缺乏必要的强归纳偏置，特别是在新语言中。遵循通用预训练和语音信息研究路线，我们提出通过在 55 种多样化语言的通用 IPA 音素和特征上进行微调，将一个单语预训练的 SSL 模型（具体而言，是在英语上训练的 HuBERT-base 模型）转变为具有强归纳偏置的通用 SSL 模型。通过直接解决先前工作的局限性，例如对大规模数据集的依赖以及缺乏显式的音系监督，我们的方法为低资源环境下的语音技术和语言记录提供了一个实用且语言导向的解决方案。我们评估了这个模型，称为 MAUBERT，在 ZRC2017 挑战赛上的表现，该挑战赛提供了 5 种训练数据少于 10h 的语言（英语、法语、德语、普通话、沃洛夫语）。为了增加评估的多样性和有效性，我们扩展了 ZRC2017 基准测试，以涵盖更多语言和场景。

5 typologically diverse languages (Swahili, Tamil, Thai, Turkish, Ukrainian). Evaluation employs the within- and across-speaker ABX metrics from ZRC2017, supplemented with metrics measuring invariance to contextual allophony (Hallap et al., 2023).

5 种类型学上差异较大的语言（斯瓦希里语、泰米尔语、泰语、土耳其语、乌克兰语）。评估使用了 ZRC2017 中的说话人内和跨说话人 ABX 指标，并补充了测量对上下文音位变体不变性的指标（Hallap 等，2023）。

Our main contributions are twofold: (i) We demonstrate that multilingual supervised fine-tuning of HuBERT for articulatory feature or phone prediction creates robust multilingual phonetic representations with strong zero-shot transfer capabilities. (ii) Our resulting models enable effective adaptation to unseen languages and casual speech with minimal self-supervised fine-tuning, achieving strong speaker and contextual invariance in new languages with only 10 h of unlabelled data. As a by-product, our method also yields candidate phoneme and feature sets for unseen languages, with potential applications for linguistic analyses of low-resource languages. The code with the data processing, methods and baselines can be found at https://github.com/bootphon/maubert.

我们的主要贡献有两个方面：(i) 我们证明了对 HuBERT 进行多语言监督微调以进行发声特征或音素预测，可以创建具有强大零样本迁移能力的稳健多语言语音表示。(ii) 我们得到的模型能够通过最小的自监督微调有效适应未见过的语言和非正式语音，在仅有 10 小时无标签数据的情况下，在新语言中实现强大的说话人和上下文不变性。作为附带产物，我们的方法还为未见过的语言生成候选音素和特征集，这在低资源语言的语言分析中具有潜在应用。数据处理、方法和基线的代码可以在 https://github.com/bootphon/maubert 找到。

2 Related Work

2 相关工作

Multilingual Speech Representation Learning. The field of multilingual speech processing has grown rapidly with large-scale semi- or self-supervised learning models that showed the potential for cross-lingual representation learning with little to no supervision (Wang et al., 2021; Conneau et al., 2021). Recent studies have expanded language coverage (Babu et al., 2022), diversified

多语言语音表征学习。随着大规模半监督或自监督学习模型的出现，多语言语音处理领域迅速发展，这些模型展示了在几乎没有监督的情况下进行跨语言表征学习的潜力（Wang 等，2021；Conneau 等，2021）。最近的研究扩大了语言覆盖范围（Babu 等，2022），并使模型更加多样化

data sources (Pratap et al., 2024), and improved efficiency (Zanon Boito et al., 2024) and robustness to noise (Chen et al., 2024b). These multilingual SSL models build upon foundational work in self-supervised speech representation learning (Baevski et al., 2020; Hsu et al., 2021; Chen et al., 2022) and have been evaluated with multi-task frameworks like SUPERB (Yang et al., 2021;

Shi et al., 2023). Meanwhile, other studies have explored the impact of phonetically-informed targets on learnt representations and their cross-lingual transferability (Wang et al., 2022; Ma et al., 2023; Feng et al., 2023). Inspired by the downstream framework of SUPERB, this work extends HuBERT-base for articulatory feature prediction.

数据来源（Pratap 等，2024），并提高了效率（Zanon Boito 等，2024）和对噪声的鲁棒性（Chen 等，2024b）。这些多语言的 SSL 模型建立在自监督语音表征学习的基础工作之上（Baevski 等，2020；Hsu 等，2021；Chen 等，2022），并且已经通过如 SUPERB（Yang 等，2021；Shi 等，2023）等多任务框架进行了评估。同时，其他研究也探讨了语音信息驱动的目标对学习表征的影响及其跨语言迁移能力（Wang 等，2022；Ma 等，2023；Feng 等，2023）。受 SUPERB 下游框架的启发，本工作将 HuBERT-base 扩展用于发音特征预测。

Articulatory Features in Speech Processing. Early work established the use of articulatory features (AFs) in speech processing (Deng and Erler, 1991; Elenius and Takacs, 1991; Eide et al., 1993), demonstrated that supervised learning can be used to automatically extract phonological features from raw (and continuous) speech (Papcun et al., 1992; King and Taylor, 2000) and produced robust articulatory/phonological feature-based speech technologies (Kirchhoff, 1999; Livescu et al., 2007; Frankel et al., 2007). The development of systematic feature inventories, particularly PanPhon (Mortensen et al., 2016), has provided practical computational tools for cross-linguistic analysis. This has enabled recent efforts to explore AFs in multilingual contexts, demonstrating their effectiveness for zero-shot multilingual speech synthesis (Staib et al., 2020) or showing their utility for cross-lingual speech recognition in low-resource languages (Feng et al., 2023).

语音处理中的发声特征。早期的研究确立了在语音处理中使用发声特征（AFs）的应用（Deng 和 Erler，1991；Elenius 和 Takacs，1991；Eide 等，1993），证明了监督学习可以用于从原始（连续）语音中自动提取语音特征（Papcun 等，1992；King 和 Taylor，2000），并产生了稳健的基于发声/语音特征的语音技术（Kirchhoff，1999；Livescu 等，2007；Frankel 等，2007）。系统性特征库存的发展，特别是 PanPhon（Mortensen 等，2016），为跨语言分析提供了实用的计算工具。这使得近期的研究能够探索在多语言环境中的 AFs，展示了它们在零样本多语言语音合成中的有效性（Staib 等，2020），或展示了它们在资源匮乏语言中的跨语言语音识别中的实用性（Feng 等，2023）。

Evaluation of Speech Representations. More specific subtasks have been developed as alternatives to downstream-based evaluation, offering clearer insights into unsupervised language learning. A prominent example is the ABX discriminability evaluation (Schatz, 2016), which assesses whether learnt representations can distinguish between different phonetic units in a way that reflects human perceptual boundaries. The Zero Resource Speech Challenge series (Versteegh et al., 2015; Dunbar et al., 2017) has systematically applied ABX evaluation to assess unsupervised speech representations, establishing benchmarks for phonetic discrimination across diverse languages and speakers. While ABX testing shows sufficient correlation with downstream performance to serve as a model comparison proxy, traditional ABX evaluation has not assessed other types of invariance, like speaking rate or speech style variations (Dunbar et al., 2022). A recent extension has begun addressing this limitation by measuring context invariance (Hallap et al., 2023). The present work builds on this extension and adds the comparison between read and casual speech.

语音表示的评估。为了替代基于下游任务的评估，已经开发了更具体的子任务，从而更清晰地了解无监督语言学习的情况。一个显著的例子是 ABX 可区分性评估（Schatz, 2016），它评估学习到的表示是否能够以反映人类感知边界的方式区分不同的语音单元。零资源语音挑战系列（Versteegh 等人，2015；Dunbar 等人，2017）已系统地应用 ABX 评估来评估无监督语音表示，为不同语言和说话人的语音辨别建立了基准。虽然 ABX 测试与下游性能之间显示出足够的相关性，可以作为模型比较的代理，但传统的 ABX 评估尚未评估其他类型的不变性，如语速或语音风格的变化（Dunbar 等人，2022）。最近的一项扩展已经开始解决这一限制，通过测量上下文不变性（Hallap 等人，2023）来应对这一问题。本工作在此基础上扩展，并增加了对朗读语音和随意语音之间比较的研究。

3 MAUBERT

3 马布尔

In this section, we introduce our Multilingual articulatory hidden-unit BERT (MAUBERT) models (Figure 1). We describe the base architecture for multilingual training (§3.1), and the self-supervised fine-tuning approach (§3.2).

在本节中，我们介绍我们的多语言发声学隐藏单元 BERT 模型（MAUBERT）（图 1）。我们描述了用于多语言训练的基本架构（§3.1），以及自监督微调方法（§3.2）。

3.1 Multilingual Pre-Training

3.1 多语言预训练

MAUBERT models are based on multilingual, continual learning of a pre-trained self-supervised speech model for articulatory feature (AF) or phone recognition (Figure 1a). We re-train HuBERT (Hsu et al., 2021) using the VoxCommunis Corpus (Ahn and Chodroff, 2022), and the associated featural annotations extracted with PanPhon (Mortensen et al., 2016).

MAUBERT 模型基于预训练的自监督语音模型的多语言持续学习，用于发音特征（AF）或音素识别（图 1a）。我们使用 VoxCommunis 数据库（Ahn 和 Chodroff，2022）对 HuBERT（Hsu 等，2021）进行再训练，并使用 PanPhon（Mortensen 等，2016）提取相关的特征注释。

We propose two versions of MAUBERT: FEAT and PHONE. The former incorporates an AF bottleneck (Figure 1a), while the latter directly predicts phones without intermediate AFs [1] .

我们提出了两种 MAUBERT 的版本：FEAT 和 PHONE。前者集成了一个 AF 瓶颈（图 1a），而后者则直接预测 phones，而不需要中间的 AF [1] 。

Encoder. We use the pre-trained HuBERT-base model as our encoder. The convolutional feature extractor is kept frozen, but the Transformer encoder is trainable. We extract the feature extractor's output after layer normalisation and dropout, as well as the outputs from each of the 12 Transformer encoder layers. The input masking is disabled during continual pre-training following SUPERB's downstream framework (Yang et al., 2021).

编码器。我们使用预训练的 HuBERT-base 模型作为我们的编码器。卷积特征提取器保持冻结状态，但 Transformer 编码器是可训练的。我们提取特征提取器在层归一化和 dropout 之后的输出，以及每个 12 个 Transformer 编码器层的输出。在遵循 SUPERB 的下游框架（Yang 等人，2021）进行持续预训练时，输入掩码功能被禁用。

Downstream modules. Preliminary experiments revealed that a simple linear layer on top of HuBERT was insufficient for the feature recognition task. Following the SUPERB framework for ASR (Yang et al., 2021), we instead leverage a weighted sum of HuBERT's intermediate representations fed into a BLSTM. Ablation experiments confirmed the importance of this contextual architecture: replacing the BLSTM with non-contextual networks

下游模块。初步实验表明，在 HuBERT 上添加一个简单的线性层对于特征识别任务是不够的。遵循 ASR 的 SUPERB 框架（Yang 等人，2021），我们转而利用 HuBERT 的中间表示的加权和，将其输入到 BLSTM 中。消融实验确认了这种上下文架构的重要性：用非上下文网络替换 BLSTM

Eval. lang.	MAUBERT variant	Feat. acc. □	Phone acc. □	PER □
Train	FEAT	95.60	72.28	30.64
PHONE	92.72	82.72	28.69	
Dev	FEAT	92.35	51.20	50.46

PHONE	88.57	67.15	48.38

评估语言	MAUBERT 变体	Feat. acc. □	手机配件 □	PER □
火车	FEAT	95.60	72.28	30.64
手机	92.72	82.72	28.69	
Dev	FEAT	92.35	51.20	50.46
手机	88.57	67.15	48.38	

Table 1: Feature and phone evaluation of MAUBERT on the held-out test set of the 55 training languages and zero-shot performance on the 5 development languages. All scores are in %.

表 1：MAUBERT 在 55 种训练语言的保留测试集上的特征和手机评估，以及在 5 种开发语言上的零样本表现。所有分数均为百分比。

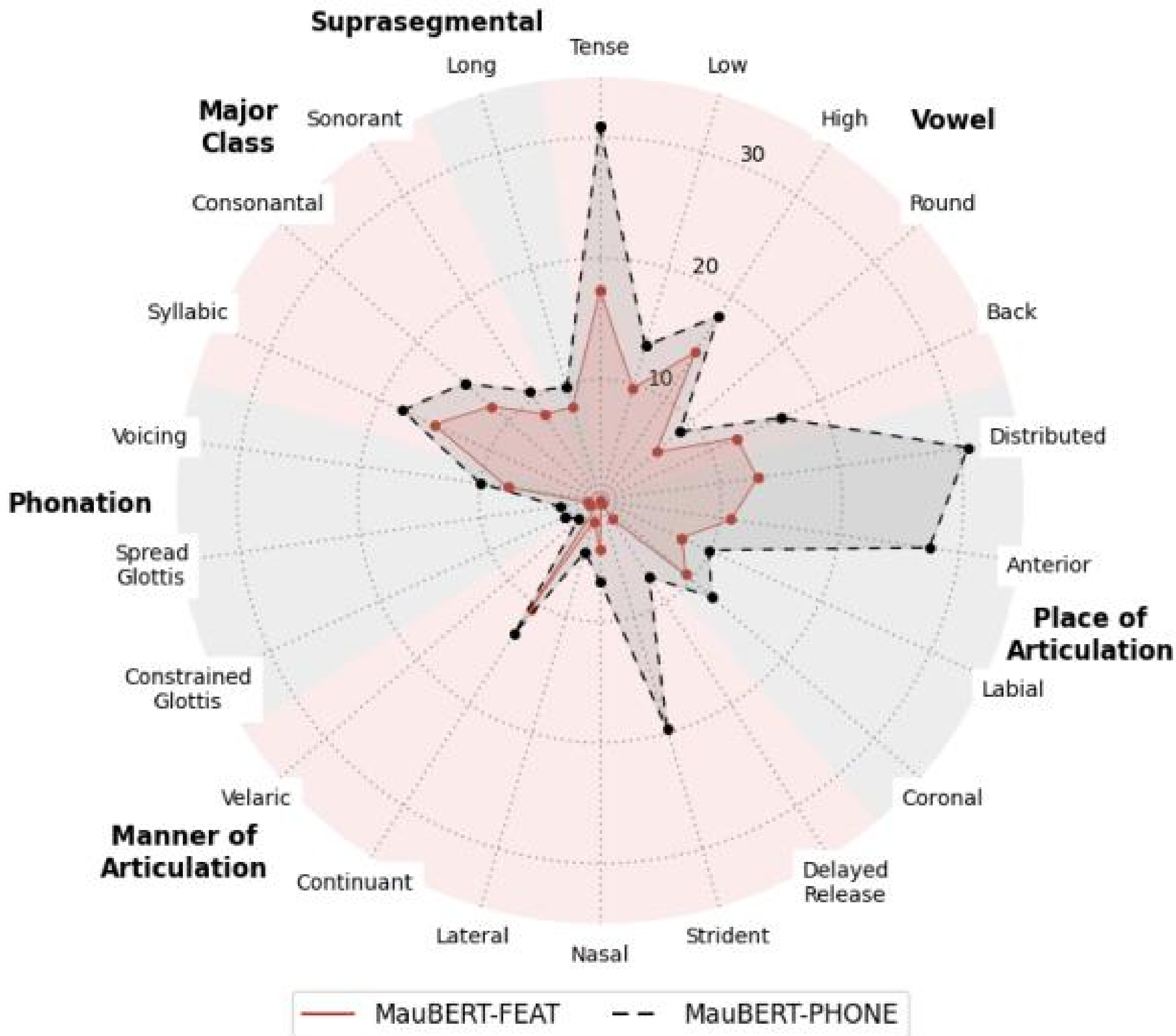

Figure 1: images/49a703b1c650a478dca8d35463adb4d6dd1ece54e29c9b863cbaf7dc0c567700.jpg

Figure 2: Classification errors across articulatory features (↓) for both MAUBERT variants on the 5 development languages. All values are in %.

图 2：两种 MAUBERT 变体在 5 种开发语言上的发声特征分类错误率（↓）。所有数值均以百分比表示。

consistently degraded performance on both the multilingual recognition tasks and the phonetic probing, motivating our adoption of the SUPERB design.

在多语言识别任务和语音探针测试中，性能持续下降，这促使我们采用了 SUPERB 设计。

Concretely, we first compute a weighted sum of the intermediate representations from our encoder and up-project them to a 1024-dimensional space. These representations are then processed through a bidirectional two-layer LSTM. Finally, we down-project the concatenated forward and backward output states into task-specific spaces: a 22-dimensional AF space for MAUBERT-FEAT and a 3293-dimensional phone space for MAUBERT-PHONE.

具体来说，我们首先计算编码器中中间表示的加权和，并将其上投影到一个 1024 维的空间中。这些表示随后通过双向两层 LSTM 进行处理。最后，我们将前向和后向输出状态连接后下投影到任务特定的空间中：对于 MAUBERT-FEAT，是一个 22 维的 AF 空间；对于 MAUBERT-PHONE，是一个 3293 维的音素空间。

Phone model. Given the non-injective nature of the feature-to-phone mapping, for MAUBERT-FEAT, we jointly learn a phone model consisting of a two-layer perceptron. Since we want the pre-training to be led by the feature recognition task only, a stop gradient operator prevents the feature hidden states from receiving any gradients from the

电话模型。鉴于特征到电话的映射具有非单射的性质，对于 MAUBERT-FEAT，我们联合学习一个由两层感知机组成的电话模型。由于我们希望预训练仅由特征识别任务引导，一个停止梯度操作符防止特征隐藏状态从任何梯度中接收来自的

Model	# Params	# Langs	# Hours	Seen dev	Seen test
MMS	965 M	1406	491 k	5	5
XEUS	577 M	4057	1 M	5	5
mHuBERT-147	95 M	147	90 k	5	4
HuBERT-base	95 M	1	960	0	1
MAUBERT (ours)	141–144 M	55	788	0	1[2]

模型	# 参数	# 语言	# 小时	已看到开发者	已测试
MMS	965 M	1406	491 k	5	5
XEUS	577 M	4057	1 M	5	5
mHuBERT-147	95 M	147	90 k	5	4
HuBERT-base	95 M	1	960	0	1
MAUBERT（我们的）	141–144 M	55	788	0	1[2]

Table 2: Comparison of speech models by number of parameters, number of languages, training data size, and development and test languages seen during training (continual learning for MAUBERT).

表 2：按参数数量、支持语言数量、训练数据规模以及训练过程中看到的开发和测试语言进行的语音模型比较（MAUBERT 的持续学习）。

phone recognition loss.

电话识别丢失。

3.2 Self-Supervised Fine-Tuning

3.2 自监督微调

We employ self-supervised fine-tuning to adapt MAUBERT models to unseen languages with limited or no labelled data. This approach generates pseudo-labels through clustering of learnt representations and applies masked language modelling (Figure 1b), enabling MAUBERT to adapt to the acoustic patterns of new languages.

我们采用自监督微调方法，将 MAUBERT 模型适应于缺乏或没有标注数据的未见过的语言。这种方法通过聚类学习到的表示生成伪标签，并应用掩码语言建模（图 1b），使 MAUBERT 能够适应新语言的声学模式。

We use four methods to generate pseudo-labels: K-means, frequent features, frequent phones and all phones. As Hsu et al. (2021), we apply K-means clustering with K = 100 to representations from encoder Transformer layers (HuBERT-base, MAUBERT) or downstream module layers (MAUBERT variants). For MAUBERT-FEAT, we extract the top K most frequent feature vectors (feat. freq.) from the articulatory feature space (top of Figure 1b). For both MAUBERT variants, we extract the top K most frequent phones (phone freq.) or all phones from pre-training data (all phones). See Appendix C for details.

我们使用四种方法生成伪标签：K-means、频繁特征、频繁音素和所有音素。如 Hsu 等人（2021）所述，我们对编码器 Transformer 层（HuBERT-base、MAUBERT）或下游模块层（MAUBERT 变体）的表示应用 K-means 聚类，其中 K = 100。对于 MAUBERT-FEAT，我们从发声特征空间（图 1b 的顶部）中提取前 K 个最频繁的特征向量（特征频率）。对于两种 MAUBERT 变体，我们从预训练数据中提取前 K 个最频繁的音素（音素频率）或所有音素（所有音素）。详情请参见附录 C。

4 MAUBERT Multilingual Pre-Training

4 MAUBERT 多语言预训练

This section describes the multilingual training and evaluation of MAUBERT variants for articulatory feature and phone recognition.

本节描述了针对发音特征和音素识别的 MAUBERT 变体的多语言训练和评估。

4.1 Data Processing

4.1 数据处理

We use the VoxCommunis Corpus, which provides phone-level annotations for a subset of Common Voice (Ardila et al., 2020) obtained with Montreal Forced Aligner (McAuliffe et al., 2017). Of the 63 covered languages, 55 are used for supervised articulatory feature prediction (totalling 788.4 h hours), 5 serve as development languages, and 3 are discarded as they were test languages. (Refer

我们使用 VoxCommunis 数据集，该数据集为 Common Voice 的一部分提供了基于音素的标注（Ardila 等，2020），该数据集是使用 Montreal Forced Aligner（McAuliffe 等，2017）获得的。在覆盖的 63 种语言中，有 55 种用于监督式发音特征预测（总计 788.4 小时），5 种作为开发语言，3 种被丢弃，因为它们是测试语言。（参考

to §5.3 for the development and test languages and to Appendix A for more data processing details.)

参见第 5.3 节了解开发和测试语言，以及参见附录 A 了解更多数据处理细节。)

Using PanPhon's feature table [3] , ternary-feature [4] annotations are derived from the phone-level annotations. Annotated segments incompatible with PanPhon are manually fixed ($e.g.$ $[t] \rightarrow [\widehat{t}], [b^{fi}] \rightarrow [b], [g] \rightarrow [g]$). Finally, we collapse the IPA table by keeping only distinct feature vectors ($e.g.$ $[\ddot{\alpha}], [e^f], [v^f], [\ddot{\alpha}]$ and $[\ddot{\alpha}]$ are all represented by the same feature vector), which reduces the table size from 6367 to 3293 segment representatives. These representatives are then used for both phone recognition and feature recognition (underlying feature values).

使用 PanPhon 的特征表 [3] ，三元特征 [4] 注释是从音素级注释中推导出来的。与 PanPhon 不兼容的注释段由人工修正（$e.g.$ $[t] \rightarrow [\widehat{t}], [b^{fi}] \rightarrow [b], [g] \rightarrow [g]$ 。最后，我们通过只保留不同的特征向量来压缩 IPA 表（$e.g.$ $[\ddot{\alpha}], [e^f], [v^f], [\ddot{\alpha}]$ 和 $[\ddot{\alpha}]$ 都由相同的特征向量表示），将表格大小从 6367 减少到 3293 个段代表。这些代表随后用于音素识别和特征识别（底层特征值）。

4.2 Training Details

4.2 训练细节

We train MAUBERT variants for feature or phone recognition across the 55 languages drawn from the VoxCommunis Corpus. Due to PanPhon's ternary feature representation, we exclude MAUBERT-FEAT predictions that correspond to zero-valued target features. Furthermore, to handle multiphthongs (hypernym of diphthong), we use a uniform heuristic so that the duration of the resulting monophthongs is roughly the same.

我们在 VoxCommunis 语料库的 55 种语言中训练 MAUBERT 的变体，用于特征或语音识别。由于 PanPhon 的三值特征表示，我们排除掉与零值目标特征相对应的 MAUBERT-FEAT 预测结果。此外，为了处理多元音（二元音的上义词），我们使用一种统一的启发式方法，使得结果中元音的持续时间大致相同。

The FEAT and PHONE variants are trained to minimise binary and multiclass cross-entropy losses, respectively, at the frame level with the Adam optimiser (Kingma and Ba, 2015). We use one V100 GPU for 40 k steps with a tri-stage learning rate schedule (4 k for warmup and 16 k for decay) that peaks at 5×10^{-5} . Following Conneau et al. (2021), we employ a language up-sampling strategy to balance the amount of data between low-resource and high-resource languages. (See Appendix A for more details.)

FEAT 和 PHONE 变体分别使用 Adam 优化器（Kingma 和 Ba，2015）在帧级别最小化二元和多类交叉熵损失。我们使用一块 V100 GPU 进行 40k 步训练，采用三阶段学习率调度策略（4k 步用于预热，16k 步用于衰减），学习率在 5×10^{-5} 达到峰值。遵循 Conneau 等人（2021）的方法，我们采用语言上采样策略来平衡低资源语言和高资源语言之间的数据量。（更多细节请参见附录 A。）

4.3 Evaluation and Results

4.3 评估与结果

We evaluate our MAUBERT models using three speech recognition metrics: frame-wise feature accuracy, frame-wise phone accuracy and phone error rate (PER) without deduplication heuristics. For feature accuracy, we compute scores over non-zero features only, excluding zero-valued target features as in training. Since MAUBERT-PHONE lacks an explicit feature space, we extract feature vectors

我们使用三个语音识别指标来评估我们的 MAUBERT 模型：帧级特征准确性、帧级音素准确性以及不使用去重启发式方法的音素错误率（PER）。对于特征准确性，我们仅在非零特征上计算得分，排除训练过程中零值的目标特征。由于 MAUBERT-PHONE 缺乏显式的特征空间，我们提取特征向量

from predicted phones using PanPhon's feature table.

使用 PanPhon 特征表预测的电话。

Table 1 shows results for both MAUBERT variants on held-out test sets from the 55 training languages and the 5 development languages. Both variants exhibit superior performance on training languages, particularly for phone-level metrics. When transitioning from training to development languages, phone accuracy drops by 15% to 21% and PER increases by approximately 20%, while feature accuracy shows more resilience with only 3–4% degradation. The FEAT variant consistently outperforms the PHONE variant in articulatory feature prediction across all languages (see Figure 2). However, this advantage does not translate to improved phone recognition performance, and the PHONE variant exhibits an even greater phone prediction advantage on development languages compared to training languages. Note that the PER gap in favour of the PHONE variant is stable across languages.

表 1 展示了两种 MAUBERT 变体在 55 种训练语言和 5 种开发语言的测试集上的结果。两种变体在训练语言上表现出色，尤其是在音素级别的指标上。当从训练语言转移到开发语言时，音素准确率下降了 15% 至 21%，而错误率（PER）增加了约 20%。相比之下，特征准确率表现出更强的鲁棒性，仅下降了 3% 至 4%。在所有语言中，FEAT 变体在发音特征预测方面始终优于 PHONE 变体（参见图 2）。然而，这种优势并未转化为更好的音素识别性能，PHONE 变体在开发语言上的音素预测优势比训练语言更大。请注意，PHONE 变体的 PER 差距在所有语言中都保持稳定。

5 Few-shot Language Adaptation

5 少样本语言适应

In this section, we assess the linguistic relevance of MAUBERT's learnt representations by evaluating their phonetic invariance across languages and speaking styles, in a zero-shot or few-shot setting.

在本节中，我们通过在零样本或少样本设置下评估其在不同语言和说话风格下的语音不变性，来评估 MAUBERT 学习到的表示的语言相关性。

5.1 Language Adaptation Setting

5.1 语言适应设置

Modes. We compare how SSL models encode speech in a new language in three modes: zero-shot, supervised fine-tuning and self-supervised fine-tuning. All the baselines and our two MAUBERT models are evaluated in zero-shot mode, while only the monolingual baseline and our two models are evaluated in the fine-tuning modes (on the 10 h training split) for fairness [5] . In the supervised mode, the models are trained to predict the ground-truth phones of masked inputs (MPR) or without masking at all (PR). In the self-supervised mode, a clustering step first produces discrete pseudo-labels, which are later used as targets for masked prediction.

模式。我们通过三种模式比较 SSL 模型如何在新语言中编码语音：零样本（zero-shot）、监督微调（supervised fine-tuning）和自监督微调（self-supervised fine-tuning）。所有基线模型和我们的两个 MAUBERT 模型均在零样本模式下进行评估，而只有单语基线模型和我们的两个模型在微调模式下进行评估（在 10 小时训练数据集上），以确保公平性 [5] 。在监督模式下，模型被训练以预测被掩码输入的真实音素（MPR），或者完全不进行掩码（PR）。在自监督模式下，首先通过聚类步骤生成离散的伪标签，这些伪标签随后被用作掩码预测的目标。

Baselines. We compare MAUBERT against several baselines, including traditional acoustic features (MFCCs), the monolingual HuBERT-base backbone, and three self-supervised models trained on massively multilingual data: MMS-1B (Pratap

基线。我们将 MAUBERT 与多个基线模型进行比较，包括传统的声学特征（MFCCs）、单语的 HuBERT-base 主干网络，以及三个在大规模多语言数据上训练的自监督模型：MMS-1B（Pratap

Systems			Development languages						Test languages (ZRC2017)							
Model	Layer	# units	triphone ABX		phoneme ABX within ctx		any ctx		avg.	1 s		triphone ABX 10 s	120 s		avg.	
WS	AS	WS	AS	WS	AS	WS	AS	WS	AS	WS	AS					
Zero-shot																
MFCC	-	39	20.00	29.00	13.23	22.36	18.05	26.33	21.49	14.78	25.58	14.70	25.33	14.70	25.32	20.07
MMS-1B	34	1280	9.37	10.74	4.76	6.02	10.53	11.37	8.80	7.58	9.02	6.91	7.91	6.91	7.83	7.69
XEUS	18	1024	6.14	7.15	3.58	4.52	9.28	9.45	6.69	4.67	5.68	4.19	4.91	4.29	4.99	4.79
mHuBERT-147	7	768	7.37	8.64	3.70	4.80	9.00	9.51	7.17	6.93	8.13	5.75	6.49	6.67	7.78	6.96
HuBERT-base	11	768	6.77	8.18	3.77	4.92	8.55	9.19	6.90	6.21	7.42	5.31	6.21	5.62	6.62	6.23
MAUBERT																
FEAT	9	768	5.49	6.52	2.95	3.81	5.97	6.47	5.20	5.86	6.84	4.78	5.57	4.86	5.68	5.60
PHONE	proj	1024	5.42	6.46	2.96	3.79	5.49	6.12	5.04	5.36	6.44	4.68	5.58	4.68	5.60	5.39
supervised FT (10 h)																
HuBERT-base																
+ PR	ws	768	4.87	6.13	2.30	3.09	3.65	4.17	4.04	5.52	6.67	4.10	4.99	4.51	5.49	5.21
+ MPR	11	768	4.26	4.98	2.05	2.62	3.94	4.30	3.69	4.26	4.84	3.25	3.73	3.89	4.36	4.05
MAUBERT																
FEAT + MPR	11	768	3.65	4.38	1.83	2.28	3.17	3.44	3.13	3.81	4.26	2.86	3.25	3.28	3.71	3.53
PHONE + MPR	12	768	3.58	4.49	1.79	2.30	2.88	3.35	3.07	3.92	4.61	2.57	3.08	2.86	3.32	3.39
self-supervised FT (10 h)																
HuBERT-base																

+ K-means (L11)	10	768	5.71	6.64	3.15	4.09	7.13	7.58	5.72	5.65	6.38	4.79	5.40	5.09	5.77	5.51
MAUBERT-FEAT																
+ K-means (L9)	10	768	4.72	5.50	2.58	3.31	5.08	5.59	4.46	5.01	5.56	4.19	4.71	4.38	5.00	4.81
+ K-means (feat)	9	768	5.00	5.81	2.69	3.41	5.29	5.69	4.65	5.16	5.92	4.30	4.98	4.51	5.20	5.01
+ feat. freq.	9	768	4.88	5.65	2.63	3.28	5.24	5.66	4.56	4.99	5.80	4.19	4.86	4.39	5.09	4.89
+ phone freq.	9	768	5.01	5.90	2.62	3.35	5.21	5.62	4.62	5.09	5.87	4.31	5.01	4.53	5.24	5.01
MAUBERT-PHONE																
+ K-means (proj)	10	768	4.91	5.71	2.66	3.32	4.93	5.55	4.51	4.84	5.62	4.17	4.81	4.38	5.15	4.83
+ K-means (phone)	10	768	4.88	5.83	2.70	3.40	5.29	5.79	4.65	5.52	6.16	4.14	4.76	4.28	4.86	4.95
+ phone freq.	10	768	4.77	5.78	2.49	3.17	4.82	5.26	4.38	5.11	5.79	4.09	4.72	4.24	4.86	4.80
+ all phones	10	768	4.88	5.84	2.49	3.16	4.85	5.28	4.42	5.15	5.89	4.05	4.70	4.20	4.83	4.80

系统			开发语言						测试语言（ZRC2017）							
模型	层	# 单位	triphone ABX □		音素 ABX □ 在语境中		任何上下文		平均值	1 s		triphone ABX □ 10 s		120 秒		平均值
WS	AS	WS	AS	WS	AS	WS	AS	WS	AS	WS	AS					
零样本																
MFCC-		39	20.00	29.00	13.23	22.36	18.05	26.33	21.49	14.78	25.58	14.70	25.33	14.70	25.32	20.07

MMS-1B	34	1280	9.37	10.74	4.76	6.02	10.53	11.37	8.80	7.58	9.02	6.91	7.91	6.91	7.83	7.69
XEUS	18	1024	6.14	7.15	3.58	4.52	9.28	9.45	6.69	4.67	5.68	4.19	4.91	4.29	4.99	4.79
mHuBERT-147	7	768	7.37	8.64	3.70	4.80	9.00	9.51	7.17	6.93	8.13	5.75	6.49	6.67	7.78	6.96
HuBERT-base	11	768	6.77	8.18	3.77	4.92	8.55	9.19	6.90	6.21	7.42	5.31	6.21	5.62	6.62	6.23
马布特																
FEAT	9	768	5.49	6.52	2.95	3.81	5.97	6.47	5.20	5.86	6.84	4.78	5.57	4.86	5.68	5.60
手机	proj	1024	5.42	6.46	2.96	3.79	5.49	6.12	5.04	5.36	6.44	4.68	5.58	4.68	5.60	5.39
监督训练(10小时)																
HuBERT-base																
+ PR	ws	768	4.87	6.13	2.30	3.09	3.65	4.17	4.04	5.52	6.67	4.10	4.99	4.51	5.49	5.21
+ MPR	11	768	4.26	4.98	2.05	2.62	3.94	4.30	3.69	4.26	4.84	3.25	3.73	3.89	4.36	4.05
马布尔																
FEAT + MPR	11	768	3.65	4.38	1.83	2.28	3.17	3.44	3.13	3.81	4.26	2.86	3.25	3.28	3.71	3.53
电话 + MPR	12	768	3.58	4.49	1.79	2.30	2.88	3.35	3.07	3.92	4.61	2.57	3.08	2.86	3.32	3.39
自监督训练(10小时)																
HuBERT-base																
+ K-means (L11)	10	768	5.71	6.64	3.15	4.09	7.13	7.58	5.72	5.65	6.38	4.79	5.40	5.09	5.77	5.51
MAUBERT-FEAT																

+ K-means (L9)	10	768	4.72	5.50	2.58	3.31	5.08	5.59	4.46	5.01	5.56	4.19	4.71	4.38	5.00	4.81
+ K-means (feat)	9	768	5.00	5.81	2.69	3.41	5.29	5.69	4.65	5.16	5.92	4.30	4.98	4.51	5.20	5.01
+ feat. 频率	9	768	4.88	5.65	2.63	3.28	5.24	5.66	4.56	4.99	5.80	4.19	4.86	4.39	5.09	4.89
+ 手机频率	9	768	5.01	5.90	2.62	3.35	5.21	5.62	4.62	5.09	5.87	4.31	5.01	4.53	5.24	5.01
MAUBERT-PHONE																
+ K-means (投影)	10	768	4.91	5.71	2.66	3.32	4.93	5.55	4.51	4.84	5.62	4.17	4.81	4.38	5.15	4.83
+ K-means (电话)	10	768	4.88	5.83	2.70	3.40	5.29	5.79	4.65	5.52	6.16	4.14	4.76	4.28	4.86	4.95
+ 手机频率	10	768	4.77	5.78	2.49	3.17	4.82	5.26	4.38	5.11	5.79	4.09	4.72	4.24	4.86	4.80
+ 所有手机	10	768	4.88	5.84	2.49	3.16	4.85	5.28	4.42	5.15	5.89	4.05	4.70	4.20	4.83	4.80

Table 3: Acoustic discriminability scores (lower is better) over 5 development languages (sw, ta, th, tr, uk) and, as test languages, the 5 languages from the Zero Resource Speech Challenge 2017 (en, fr, zh, de, wo). The best layer for each model is selected based on the average ABX score on the development languages. The best scores are in bold and the second best are underlined.

表 3：在 5 种开发语言（sw, ta, th, tr, uk）和 5 种测试语言（来自 2017 年零资源语音挑战赛的 en, fr, zh, de, wo）上的声学可区分性得分（得分越低越好）。每个模型的最佳层是根据在开发语言上的平均 ABX 得分选择的。最佳得分以粗体显示，次佳得分以下划线显示。

et al., 2024), mHuBERT-147 (Zanon Boito et al., 2024), and XEUS (Chen et al., 2024b). Table 2 shows a brief comparison of the training data between the baselines and our models.

et al., 2024), mHuBERT-147 (Zanon Boito et al., 2024), 和 XEUS (Chen et al., 2024b)。表 2 展示了基线模型与我们的模型之间训练数据的简要比较。

Implementation. For the supervised fine-tuning, we train the models for 20 k steps on one V100 GPU with a tri-stage learning rate schedule (2 k for warmup and 8 k for decay). We use the Adam optimiser with a peak learning rate at 1×10^{-4} . For the self-supervised fine-tuning, we train the Transformer encoder for 50 k steps on one H100 GPU. We use the Adam optimiser with a linear decay schedule (8 % for warmup, then linear decay back to zero) that peaks at 5×10^{-6} .

实现。对于监督微调，我们在一个 V100 GPU 上使用三阶段学习率调度方案训练模型 20k 步（2k 用于预热，8k 用于衰减）。我们使用 Adam 优化器，峰值学习率为 1×10^{-4} 。对于自监督微调，我们在一个 H100 GPU 上训练 Transformer 编码器 50k 步。我们使用 Adam 优化器，采用线性衰减调度方案（预热阶段为 8%，然后线性衰减至零），峰值学习率为 5×10^{-6} 。

5.2 Metric

5.2 指标

We employ the ABX discriminability test to measure phonetic invariance (Schatz, 2016). It evaluates speech representations by comparing distances between three triphones: A, X (same linguistic unit as A), and B (different unit). The test is considered successful when the distance between A and X is smaller than that between A and B. The test com-

我们采用 ABX 辨别性测试来衡量语音不变性（Schatz, 2016）。该测试通过比较三个三音素（triphone）之间的距离来评估语音表示：A、X（与 A 相同的语言单位）和 B（不同的单位）。当 A 与 X 之间的距离小于 A 与 B 之间的距离时，该测试被认为是成功的。该测试继续

prises two variants: a triphone-based version that examines complete triphone representations, and a phoneme-based version that focuses exclusively on central phone representations.

采用两种变体：一种是基于三音素的版本，考察完整的三音素表示；另一种是基于音素的版本，仅专注于中心音素表示。

The speaker condition varies between two scenarios: within-speaker (all triphones share the speaker) and across-speaker (only A and B share the speaker). In addition, contextual conditions across all three items (A, B, and X) can be manipulated: within-context (where all items share identical surrounding phonetic context) versus any-context (where surrounding contexts may differ).

说话人条件在两种情境下有所不同：同说话人条件（所有三音素共享同一说话人）和跨说话人条件（只有 A 和 B 共享同一说话人）。此外，还可以操控所有三个项目（A、B 和 X）的上下文条件：同上下文条件（所有项目共享相同的周围语音上下文）与任意上下文条件（周围上下文可能不同）。

We compute all the ABX scores with the CPU backend of fastabx (Poli et al., 2025a).

我们使用 fastabx（Poli 等人，2025a）的 CPU 后端计算所有 ABX 分数。

5.3 Language Data

5.3 语言数据

Following the Zero Resource Speech Challenge 2017 (Dunbar et al., 2017), we curate ABX-ready datasets for five development languages from the VoxCommunis Corpus: Swahili, Tamil, Thai, Turkish and Ukrainian. The ABX datasets consist of three splits for each language: a 10 h training set, a validation set and a test set. We select the best parameters, hyperparameters and layers of the various

在 2017 年零资源语音挑战赛（Dunbar 等，2017）之后，我们从 VoxCommunis 语料库中为五种开发语言整理了 ABX 准备好的数据集：斯瓦希里语、泰米尔语、泰语、土耳其语和乌克兰语。每个语言的 ABX 数据集包含三个子集：一个 10 小时的训练集、一个验证集和一个测试集。我们选择各种模型的最佳参数、超参数和层。

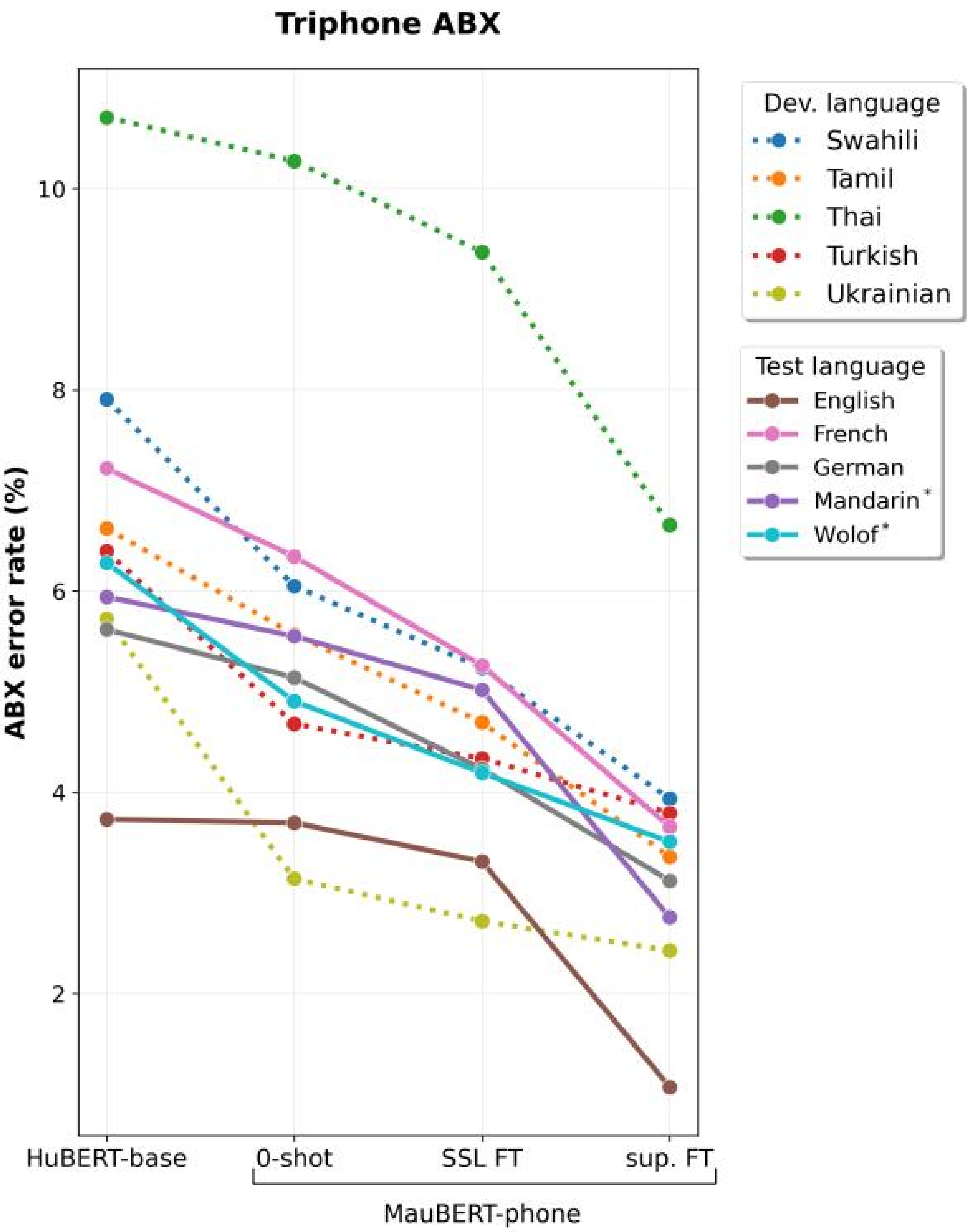

Figure 2: images/e2c2a257cbfbb307e664dde8e17559fed60568ae2328a5fd0c9b7f1f6672f270.jpg

Figure 3: Reduction of the triphone ABX error rates across the 5 development languages and 5 test languages between the base HuBERT model, and MAUBERT, tested in zero-shot and after masked fine-tuning (10 h) with or without labels on a new language. The two speaker conditions are averaged, and the 10 s subset is chosen for the test languages. *Mandarin and Wolof only have 1.5 and 1.8 h of training data, resp.

图 3：在零样本设置和在新语言上进行掩码微调（10 小时）且有或没有标签的情况下，基线 HuBERT 模型与 MAUBERT 在 5 种开发语言和 5 种测试语言之间对三音素 ABX 错误率的减少情况。两种说话人条件的平均值被计算，测试语言的 10 秒子集被选择。* 汉语和沃洛夫语分别只有 1.5 和 1.8 小时的训练数据。

models according to their impact on the average ABX score (triphone-based ABX, within-context phoneme ABX and any-context phoneme ABX) on the ABX test sets.

根据它们对平均 ABX 分数的影响来评估模型（基于三音素的 ABX、上下文内的音素 ABX 和任意上下文音素 ABX）在 ABX 测试集上的表现。

We use both the development and surprise languages from the aforementioned Zero Resource Speech Challenge 2017, namely English, French, Mandarin, German and Wolof, as test languages (hereafter referred to as ZRC2017). The amount of speech in the original training set ranges from 2.3 h for Mandarin to 35.3 h for English. We thus extract training and validation splits of up to 10 h. In line with the desiderata of the challenge, we keep the original test subsets of differing length (1 s, 10 s and 120 s) to evaluate the effect of context length (triphone-based ABX only). We evaluate only the best configuration for each model on these languages.

我们使用上述零资源语音挑战 2017（ZRC2017）中提到的开发和惊喜语言，即英语、法语、普通话、德语和沃洛夫语，作为测试语言。原始训练集中的语音量从普通话的 2.3 小时到英语的 35.3 小时不等。因此，我们提取最多 10 小时的训练和验证子集。按照挑战的要求，我们保留了不同长度的原始测试子集（1 秒、10 秒和 120 秒），以评估上下文长度（基于三音素的 ABX）的影响。我们仅在这些语言上评估每个模型的最佳配置。

Additionally, we curate an ABX dataset of casual speech in English and French, sourcing high-quality recorded conversations of native speakers. The dataset possesses the same three-split structure as the development languages.

此外，我们还整理了一个包含英语和法语口语的 ABX 数据集，该数据集来源于母语者的高质量录音对话。该数据集具有与开发语言相同的三折结构。

Systems	Read		Casual	
WS	AS	WS	AS	
MMS	6.75	8.99	13.47	17.47
XEUS	4.46	5.78	8.69	11.29
mHuBERT-147	5.29	6.83	10.62	13.72
HuBERT-base	4.71	6.24	9.41	12.48
ours				
MAUBERT-FEAT	4.45	5.75	9.43	12.11
+ K-means (L9)	3.95	5.00	8.25	10.66
MAUBERT-PHONE	4.29	5.75	9.23	11.92
+ phone freq.	3.69	4.89	8.43	10.85

系统	阅读		休闲	
WS	AS	WS	AS	
MMS	6.75	8.99	13.47	17.47
XEUS	4.46	5.78	8.69	11.29
mHuBERT-147	5.29	6.83	10.62	13.72
HuBERT-base	4.71	6.24	9.41	12.48
我们的				

MAUBERT-FEAT	4.45	5.75	9.43	12.11
+ K-means (L9)	3.95	5.00	8.25	10.66
MAUBERT-PHONE	4.29	5.75	9.23	11.92
+ 电话频率	3.69	4.89	8.43	10.85

Table 4: Triphone-based ABX error rates across registers (read vs. spontaneous) for English and French in zero-shot mode. Our two MAUBERT variants are also tested after self-supervised fine-tuning on 10 h.

表 4：在零样本模式下，英语和法语的三音素基 ABX 错误率在不同注册（朗读 vs. 自然）中的表现。我们的两个 MAUBERT 变体也在 10 小时自监督微调后进行了测试。

5.4 Few-shot Language Adaptation Results

5.4 少样本语言适应结果

Our experimental results demonstrate the competitive performance of our MAUBERT models across multiple evaluation scenarios.

我们的实验结果表明，我们的 MAUBERT 模型在多个评估场景中表现出竞争力。

Multilingual Training Benefits. The top of Table 3 illustrates the phonetic invariance performance in zero-shot mode achieved by MAUBERT models through multilingual pre-training. Our models attain particularly strong results in the any-context phoneme-based ABX tasks, and the MAUBERT-PHONE model delivers the best overall zero-shot performance (5.22% against 5.74% for XEUS). Further, Figure 3 confirms the performance improvements of the multilingual pretraining as well as the proposed self-supervised fine-tuning: 6.62% for the HuBERT-base baseline vs. 5.54% for MAUBERT-PHONE in zero-shot and 4.84% after phone freq. MPR in triphone ABX with similar audio lengths.

多语言训练的优势。表 3 顶部展示了通过多语言预训练，MAUBERT 模型在零样本模式下实现的语音不变性性能。我们的模型在任意上下文的基于音素的 ABX 任务中表现出特别强的性能，而 MAUBERT-PHONE 模型在整体零样本性能上表现最佳（5.22%，相比之下 XEUS 为 5.74%）。此外，图 3 确认了多语言预训练以及所提出的自监督微调带来的性能提升：在零样本条件下，HuBERT-base 基线模型的性能为 6.62%，而 MAUBERT-PHONE 模型为 5.54%；在音素频率 MPR 处理后，三音素 ABX 任务中，音频长度相似的情况下，性能进一步提升至 4.84%。

Cross-linguistic Performance Patterns. Table 3 also shows that development languages present greater challenges than test languages when evaluated under comparable conditions (triphone ABX with similar audio lengths) across models and modes. This pattern indicates varying degrees of phonetic complexity across language families and suggests that our model selection strategy (detailed in §5.3) based on development languages' performance provides a robust foundation for cross-lingual generalisation. Figure 3 reinforces this observation, with development languages (shown with dotted lines) generally exhibiting higher error rates and more variable performance across the

跨语言表现模式。表 3 还显示，在相同条件下（三音素 ABX 测试，音频长度相似）评估模型和模式时，开发语言比测试语言带来更大的挑战。这种模式表明不同语言家族在语音复杂性方面存在不同程度的差异，并表明我们基于开发语言表现的模型选择策略（详见第 5.3 节）为跨语言泛化提供了稳固的基础。图 3 进一步支持了这一观察，开发语言（用虚线表示）通常表现出更高的错误率，并且在不同测试中表现更加不稳定。

training progression compared to test languages (solid lines), suggesting the former present more challenging phonetic discrimination tasks and may represent more diverse or complex phonological systems.

训练进展与测试语言（实线）相比，表明前者具有更具挑战性的语音辨别任务，可能代表更加多样化或复杂的语音系统。

Supervised Fine-tuning Efficacy. Supervised fine-tuning yields substantial improvements in ABX error rates. Particularly striking is the effectiveness of predicting the ground-truth phones of masked inputs (MPR), which reduces ABX error rates compared to standard phone prediction (PR), especially for triphone-based ABX. The MAUBERT-PHONE + MPR configuration achieves the best supervised performance (3.07% on development languages, 3.39% on test languages), representing a significant 38% relative improvement over the zero-shot baseline. Figure 3 illustrates this systematic improvement pattern across all languages, with supervised fine-tuning showing the most notable gains (3.43% average ABX score). Remarkably, fine-tuning effectiveness appears largely independent of training data quantity: low-resource language Wolof achieves comparable error rates to high-resource languages, indicating robust few-shot adaptation capabilities.

监督微调效果。监督微调在 ABX 错误率方面带来了显著的改进。特别引人注目的是，预测被掩码输入的真实音素（MPR）的有效性，这比标准音素预测（PR）在 ABX 错误率上有所降低，尤其是在基于三音素的 ABX 任务中表现尤为突出。MAUBERT-PHONE + MPR 配置在监督性能方面表现最佳（开发语言上为 3.07%，测试语言上为 3.39%），相比零样本基线实现了显著的 38% 相对提升。图 3 展示了所有语言中这种系统性改进模式，监督微调显示出最显著的提升（平均 ABX 得分为 3.43%）。值得注意的是，微调效果似乎与训练数据量的多少关系不大：资源匮乏的语言沃洛语实现了与资源丰富的语言相当的错误率，这表明其具备强大的少样本适应能力。

Self-supervised Fine-tuning Analysis. While self-supervised fine-tuning approaches show consistent improvements over zero-shot performance, a performance gap remains compared to the fully-supervised standard. Among the clustering strategies, our phone frequency-based approach demonstrates some gains over standard K-means clustering, particularly excelling in phoneme-level discrimination tasks and longer temporal contexts (10 s and 120 s triphone ABX). MAUBERT-PHONE with phone frequency clustering achieves the best self-supervised performance (4.59% average ABX score), highlighting the value of linguistically-informed clustering strategies.

自监督微调分析。虽然自监督微调方法在零样本性能上表现出持续的改进，但与完全监督的标准相比，性能差距仍然存在。在聚类策略中，我们的基于音素频率的方法在标准 K-means 聚类的基础上取得了一些提升，尤其是在音素级区分任务和更长的时序上下文（10 秒和 120 秒三音素 ABX）中表现尤为突出。采用音素频率聚类的 MAUBERT-PHONE 在自监督性能方面达到最佳水平（平均 ABX 得分为 4.59%），突显了语言信息驱动的聚类策略的价值。

Speech Register Adaptation Results. Table 4 reveals nuanced domain-specific patterns across read versus casual speech. In zero-shot mode, our models perform slightly better than multilingual baselines on read speech (MAUBERT-PHONE: 5.02% vs. XEUS: 5.12%) but show reversed performance on casual speech (10.58% vs. 9.99% for XEUS), reflecting the inherent difficulty of spontaneous speech processing with its increased phonetic variability and reduced articulatory precision. However, self-supervised fine-tuning not only amplifies our advantage on read speech (4.29%) but also recovers competitive performance on casual speech (9.64%), demonstrating the robustness of our adaptation approach across speech domains.

语音注册适应结果。表 4 揭示了在朗读与随意性语音之间存在细微的领域特定模式。在零样本模式下，我们的模型在朗读语音上表现略优于多语言基线模型（MAUBERT-PHONE: 5.02% vs. XEUS: 5.12%），但在随意性语音上表现则相反（10.58% vs. 9.99% 对于 XEUS），这反映了自发性语音处理的固有困难，其语音变化性增加且发音精确度降低。然而，自监督微调不仅增强了我们在朗读语音上的优势（4.29%），还恢复了在随意性语音上的竞争力表现（9.64%），展示了我们的适应方法在语音领域上的鲁棒性。

5.5 Phonetic Inventory Discovery Results

5.5 发音系统调查结果

Two of the MAUBERT SSL methods consist in assigning a feature or a phoneme set to a new language as a target for SSL fine-tuning. These methods amount to discovering the phonetic inventories of previously unseen languages [6] . Following Želasko et al. (2022), we leverage the frequency distribution of (discrete) articulatory feature vectors produced by MAUBERT-FEAT, where high-frequency combinations likely correspond to actual phones in the language inventory [7] .

两种 MAUBERT SSL 方法包括将一个特征或一个音素集分配给一种新语言，作为 SSL 微调的目标。这些方法相当于发现之前未见过的语言的音素库存 [6] 。根据 Želasko 等人（2022）的研究，我们利用 MAUBERT-FEAT 产生的（离散）发音特征向量的频率分布，其中高频组合很可能对应于语言库存中的实际音素 [7] 。

Table 7 reveals a clear trade-off between precision and recall across different threshold strategies. The top-100 approach achieves consistently high recall (at least 0.825 for four out of five languages), successfully capturing most phonemes in the target inventories. However, this comes at the cost of precision (0.270–0.390), indicating substantial inclusion of spurious feature vectors. Conversely, the optimised frequency threshold approach significantly improves precision (0.778–0.872) while maintaining reasonable recall (0.532–0.810), suggesting more accurate phonetic identification with fewer false positives.

表 7 揭示了不同阈值策略在精确率和召回率之间存在明显的权衡。top-100 方法实现了持续的高召回率（五种语言中有四种至少为 0.825），成功地捕捉了目标音系中的大多数音素。然而，这种做法是以牺牲精确率（0.270–0.390）为代价的，表明包含了大量虚假的特征向量。相反，优化后的频率阈值方法显著提高了精确率（0.778–0.872），同时保持了合理的召回率（0.532–0.810），表明在减少误报的情况下实现了更准确的音素识别。

The superior F_1 performance of optimised thresholds over fixed thresholds underscores the importance of adaptive, data-driven approaches to inventory discovery. (See Table 8 for some inventory examples with F_1 -optimal thresholds.)

优化后的阈值相比固定阈值表现出更优越的 F_1 性能，这突显了采用适应性、数据驱动方法进行库存发现的重要性。(参见表 8，其中包含一些带有 F_1 -最优阈值的库存示例。)

6 Discussion

6 讨论

Broader Impact. Our demonstration that effective phonetic models can be developed for low-resource languages with minimal training data (as evidenced by Wolof performance with less than 2 h of data) is an encouraging signal towards more linguistic inclusion in computational models. In addition, our frequency-based methodology offers particular value for endangered language documentation, where traditional phonological analysis may

更广泛的影响。我们展示出，可以使用极少的训练数据（如沃洛夫语性能仅需不到 2 小时的数据）为低资源语言开发出有效的语音模型，这为计算模型中实现更多的语言包容性提供了积极的信号。此外，我们的基于频率的方法对于濒危语言的记录具有特别的价值，在这种情况下，传统的音系分析可能

be impractical, providing linguists with not only a multilingual articulatory feature recogniser but also an automated tool for initial phonetic hypothesis generation that can guide subsequent detailed analysis. However, the superior performance of high-resource languages like English also highlights the importance of linguistic diversity in training data, since the imbalance thereof could persist through evaluation.

不切实际，为语言学家提供了一个多语言发声特征识别器，同时也提供了一个用于初步语音假设生成的自动化工具，以指导后续的详细分析。然而，像英语这样的高资源语言的卓越表现也突显了训练数据中语言多样性的的重要性，因为这种不平衡可能会在评估过程中持续存在。

Future Work. Several promising research directions emerge from our findings. The counterintuitive relationship between training data quantity and fine-tuning effectiveness suggests that investigation into optimal data selection strategies could yield significant improvements, potentially focusing on phonetically diverse rather than simply large datasets. Given that speaker and content information are encoded at different layers of HubERT's encoder, selectively targeting a subset of hidden layers during multilingual pre-training is another promising avenue worth exploring. The domain adaptation capabilities demonstrated in our casual speech experiments indicate potential for developing more robust models through multidomain training paradigms. Furthermore, extending the self-supervised fine-tuning beyond the encoder to encompass the entire MAUBERT architecture could address current limitations by enabling end-to-end adaptation of both the pre-trained representations and the downstream articulatory feature prediction modules, potentially leading to improved performance on target languages and domains, and better phonetic inventory discovery.

未来工作。从我们的研究结果中出现了几个有前景的研究方向。训练数据量与微调效果之间的反直觉关系表明，对最优数据选择策略进行研究可能会带来显著的改进，可能更关注语音多样性而非仅仅数据量大的数据集。鉴于说话人和内容信息在 HubERT 编码器的不同层中进行编码，在多语言预训练过程中选择性地针对一部分隐藏层进行训练是另一个值得探索的有前景方向。我们在随意语音实验中展示的领域适应能力表明，通过多领域训练范式可以开发出更加鲁棒的模型。此外，将自监督微调扩展到编码器之外，涵盖整个 MAUBERT 架构，可以通过使预训练表示和下游发音特征预测模块都能进行端到端的适应，从而解决当前的局限性，可能在目标语言和领域上实现更好的性能，并更好地发现语音库存。

7 Conclusion

7 结论

This work presents MAUBERT, a multilingual extension of HuBERT that demonstrates competitive phonetic discrimination capabilities across diverse languages while revealing important insights about cross-lingual representation learning. Our results establish that multilingual supervised pre-training creates robust phonetic foundations that enable effective few-shot adaptation to new languages (10 hours of speech) with or without supervision. The demonstrated effectiveness on both read and spontaneous speech, coupled with strong performance on low-resource languages, positions this work as a significant step towards more inclusive multilingual speech technologies.

这项工作提出了 MAUBERT，这是 HuBERT 的多语言扩展，它在多种语言中展示了具有竞争力的语音辨别能力，同时揭示了关于跨语言表示学习的重要见解。我们的结果表明，多语言监督预训练能够创建稳健的语音基础，从而使得在有或没有监督的情况下，能够有效地对新语言（10 小时的语音）进行少量样本适应。在读出语音和自发语音上展示的有效性，以及在资源匮乏语言上的强性能，使这项工作成为迈向更加包容的多语言语音技术的重要一步。

Limitations

限制

The evaluation is constrained to the ABX discrimination task, which, while established as a standard phonetic benchmark, may not fully capture the nuanced linguistic representations as required for other linguistic levels (e.g. syntax and semantics). The performance gap between self-supervised and supervised fine-tuning methods suggests that clustering- or frequency-based approaches, despite their linguistic motivation, remain suboptimal compared to gold-standard supervision.

评估仅限于 ABX 辨别任务，尽管该任务已被确立为语音学的标准基准，但它可能无法充分捕捉其他语言层级（如句法和语义）所需的细微语言表示。自监督与监督微调方法之间的性能差距表明，尽管基于聚类或频率的方法具有语言学动机，但它们仍不如黄金标准监督方法优越。

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References

参考资料

Emily Ahn and Eleanor Chodroff. 2022. VoxCommunis: A corpus for cross-linguistic phonetic analysis. In Proceedings of the Thirteenth Language Resources and Evaluation Conference, pages 5286–5294, Marseille, France. European Language Resources Association.

Rosana Ardila, Megan Branson, Kelly Davis, Michael Kohler, Josh Meyer, Michael Henretty, Reuben Morais, Lindsay Saunders, Francis Tyers, and Gregor Weber. 2020. Common Voice: A massively-multilingual speech corpus. In Proceedings of the Twelfth Language Resources and Evaluation Conference, pages 4218–4222, Marseille, France. European Language Resources Association.

Arun Babu, Changhan Wang, Andros Tjandra, Kushal Lakhotia, Qiantong Xu, Naman Goyal, Kritika Singh, Patrick von Platen, Yatharth Saraf, Juan Pino, Alexei Baevski, Alexis Conneau, and Michael Auli. 2022. XLS-R: Self-supervised cross-lingual speech representation learning at scale. In Interspeech 2022, pages 2278–2282.

Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli. 2020. wav2vec 2.0: A framework for self-supervised learning of speech representations. In Advances in Neural Information Processing Systems, volume 33, pages 12449–12460. Curran Associates, Inc.

Can Balioglu, Alexander Erben, Martin Gleize, Artyom Kozhevnikov, Ilia Kulikov, and Julien Yao. 2023. fairseq2.

Zalán Borsos, Raphaël Marinier, Damien Vincent, Eugene Kharitonov, Olivier Pietquin, Matt Sharifi, Dominik Roblek, Olivier Teboul, David Grangier, Marco Tagliasacchi, and Neil Zeghidour. 2023. Audiolm: A language modeling approach to audio generation. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 31:2523–2533.

Chih-Chen Chen, William Chen, Rodolfo Joel Zevallos, and John E Ortega. 2024a. Evaluating self-supervised speech representations for indigenous American languages. In Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024), pages 6444–6450.

Peng-Jen Chen, Kevin Tran, Yilin Yang, Jingfei Du, Justine Kao, Yu-An Chung, Paden Tomasello, Paul-Ambroise Duquenne, Holger Schwenk, Hongyu Gong, Hirofumi Inaguma, Sravya Popuri, Changhan Wang, Juan Pino, Wei-Ning Hsu, and Ann Lee. 2023. Speech-to-speech translation for a real-world unwritten language. In Findings of the Association for Computational Linguistics: ACL 2023, pages 4969–4983, Toronto, Canada. Association for Computational Linguistics.

Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Jian Wu, Michael Zeng, Xiangzhan Yu, and Furu Wei. 2022. WavLM: Large-scale self-supervised pre-training for full stack speech processing. IEEE Journal of Selected Topics in Signal Processing, 16(6):1505–1518.

William Chen, Wangyou Zhang, Yifan Peng, Xinjian Li, Jinchuan Tian, Jiatong Shi, Xuankai Chang, Soumi Maiti, Karen Livescu, and Shinji Watanabe. 2024b. Towards robust speech representation learning for thousands of languages. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pages 10205–10224, Miami, Florida, USA. Association for Computational Linguistics.

Alexis Conneau, Alexei Baevski, Ronan Collobert, Abdelrahman Mohamed, and Michael Auli. 2021. Unsupervised cross-lingual representation learning for speech recognition. In Interspeech 2021, pages 2426–2430.

Maureen de Seyssel, Marvin Lavechin, Yossi Adi, Emmanuel Dupoux, and Guillaume Wisniewski. 2022. Probing phoneme, language and speaker information in unsupervised speech representations. In Interspeech 2022, pages 1402–1406.

Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. 2024. Moshi: A speech-text foundation model for real-time dialogue. Preprint, arXiv:2410.00037. Version 2.

Li Deng and Kevin Erler. 1991. Microstructural speech units and their hmm representation for discrete utterance speech recognition. In [Proceedings] ICASSP 91: 1991 International Conference on Acoustics, Speech, and Signal Processing, pages 193–196 vol.1.

Ewan Dunbar, Xuan Nga Cao, Juan Benjumea, Julien Karadayi, Mathieu Bernard, Laurent Besacier, Xavier Anguera, and Emmanuel Dupoux. 2017. The zero resource speech challenge 2017. In 2017 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU), pages 323–330.

Ewan Dunbar, Nicolas Hamilakis, and Emmanuel Dupoux. 2022. Self-supervised language learning from raw audio: Lessons from the zero resource speech challenge. IEEE Journal of Selected Topics in Signal Processing, 16(6):1211–1226.

Ellen Eide, Jan Robin Rohlicek, Herbert Gish, and Sanjoy Mitter. 1993. A linguistic feature representation of the speech waveform. In 1993 IEEE International Conference on Acoustics, Speech, and Signal Processing, volume 2, pages 483–486 vol.2.

Kjell Elenius and Gy Takacs. 1991. Phoneme recognition with an artificial neural network. In 2nd European Conference on Speech Communication and Technology (Eurospeech 1991), pages 121–124.

Emily Ahn 和 Eleanor Chodroff. 2022. VoxCommunis: 用于跨语言语音分析的语料库. 在第十三届语言资源与评价会议 (LREC) 会议论文集, 第 5286–5294 页, 法国马赛. 欧洲语言资源协会.

Rosana Ardila, Megan Branson, Kelly Davis, Michael Kohler, Josh Meyer, Michael Henretty, Reuben Morais, Lindsay Saunders, Francis Tyers, 和 Gregor Weber. 2020. Common Voice: 一个大规模多语言语音语料库. 在第十二届语言资源与评价会议 (LREC) 会议论文集, 第 4218–4222 页, 法国马赛. 欧洲语言资源协会.

Arun Babu, Changhan Wang, Andros Tjandra, Kushal Lakhotia, Qiantong Xu, Naman Goyal, Kritika Singh, Patrick von Platen, Yatharth Saraf, Juan Pino, Alexei Baevski, Alexis Conneau, 和 Michael Auli. 2022. XLS-R: 大规模跨语言语音表示的自监督学习. 在 2022 年国际语音研讨会 (Interspeech 2022), 第 2278–2282 页.

Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, 和 Michael Auli. 2020. wav2vec 2.0: 语音表示的自监督学习框架. 在神经信息处理系统大会 (NeurIPS) 2020, 卷 33, 第 12449–12460 页. Curran Associates, Inc.

Can Balioglu, Alexander Erben, Martin Gleize, Artyom Kozhevnikov, Ilia Kulikov, 和 Julien Yao. 2023. fairseq2.

Zalán Borsos, Raphaël Marinier, Damien Vincent, Eugene Kharitonov, Olivier Pietquin, Matt Sharifi, Dominik Roblek, Olivier Teboul, David Grangier, Marco Tagliasacchi, 和 Neil Zeghidour. 2023. Audiolm: 音频生成的语言建模方法. IEEE/ACM 音频、语音与语言处理汇刊, 31:2523–2533.

Chih-Chen Chen, William Chen, Rodolfo Joel Zevallos, 和 John E Ortega. 2024a. 评估用于美洲原住

民语言的自监督语音表示. 在 2024 年联合国际计算语言学、语言资源与评价会议 (LREC-COLING 2024) 会议论文集, 第 6444–6450 页.
Peng-Jen Chen, Kevin Tran, Yilin Yang, Jingfei Du, Justine Kao, Yu-An Chung, Paden Tomasello, Paul-Ambroise Duquenne, Holger Schwenk, Hongyu Gong, Hirofumi Inaguma, Sravya Popuri, Changhan Wang, Juan Pino, Wei-Ning Hsu, 和 Ann Lee. 2023. 实际未书写语言的语音到语音翻译. 在计算语言学协会 (ACL) 2023 年会议论文集, 第 4969–4983 页, 加拿大多伦多. 计算语言学协会.
Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Jian Wu, Michael Zeng, Xiangzhan Yu, 和 Furu Wei. 2022. WavLM: 用于完整语音处理的大型自监督预训练. IEEE 选定主题在信号处理汇刊, 16(6):1505–1518.
William Chen, Wangyou Zhang, Yifan Peng, Xinjian Li, Jinchuan Tian, Jiatong Shi, Xuankai Chang, Soumi Maiti, Karen Livescu, 和 Shinji Watanabe. 2024b. 向数千种语言的鲁棒语音表示学习迈进. 在 2024 年自然语言处理经验方法会议 (EMNLP 2024) 会议论文集, 第 10205–10224 页, 美国佛罗里达州迈阿密. 计算语言学协会.
Alexis Conneau, Alexei Baevski, Ronan Collobert, Abdelrahman Mohamed, 和 Michael Auli. 2021. 语音识别的无监督跨语言表示学习. 在 2021 年国际语音研讨会 (Interspeech 2021), 第 2426–2430 页.
Maureen de Seyssel, Marvin Lavechin, Yossi Adi, Emmanuel Dupoux, 和 Guillaume Wisniewski. 2022. 在无监督语音表示中探测音素、语言和说话人信息. 在 2022 年国际语音研讨会 (Interspeech 2022), 第 1402–1406 页.
Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, 和 Neil Zeghidour. 2024. Moshi: 实时对话的语音-文本基础模型. 预印本, arXiv:2410.00037. 版本 2.
Li Deng 和 Kevin Erler. 1991. 微结构语音单元及其 HMM 表示用于离散语音识别. 在 [会议论文集] ICASSP 91: 1991 国际声学、语音与信号处理会议, 第 193–196 页, 卷 1.
Ewan Dunbar, Xuan Nga Cao, Juan Benjumea, Julien Karadayi, Mathieu Bernard, Laurent Besacier, Xavier Anguera, 和 Emmanuel Dupoux. 2017. 2017 年零资源语音挑战. 在 2017 年 IEEE 自动语音识别与理解研讨会 (ASRU), 第 323–330 页.
Ewan Dunbar, Nicolas Hamilakis, 和 Emmanuel Dupoux. 2022. 从原始音频中进行自监督语言学习: 零资源语音挑战的启示. IEEE 选定主题在信号处理汇刊, 16(6):1211–1226.
Ellen Eide, Jan Robin Rohlicek, Herbert Gish, 和 Sanjoy Mitter. 1993. 语音波形的语言特征表示. 在 1993 年 IEEE 国际声学、语音与信号处理会议, 卷 2, 第 483–486 页, 卷 2.
Kjell Elenius 和 Gy Takacs. 1991. 使用人工神经网络进行音素识别. 在第二届欧洲语音通信与技术会议 (Eurospeech 1991), 第 121–124 页.